Lathi 20题练习册 第11批

对应 Boulet 前四章

原书英文题干、公式与必要图示 · 无答案版

记号：D = d/dt；差分式中 E 为前移算子，E{y[n]} = y[n+1]；j^2 = -1
u 为单位阶跃，δ 为单位冲激；LTIC / LTID 分别指连续 / 离散时间 LTI 系统

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来源：B. P. Lathi / R. A. Green, Linear Systems and Signals, 3rd ed. (2018)
中文主题仅用于课程对应；题目要求以原书英文为准。

开关间隔递增的二进制信号：能量与功率

Lathi 1.1-12 | 原书第 138、139 页

Boulet 第1章：Finite-Energy and Finite-Power Signals, pp.21-23

1.1-12 A binary signal $x(t) = 0$ for $t < 0$. For positive time, $x(t)$ toggles between one and zero as follows: one for 1 second, zero for 1 second, one for 1 second, zero for 2 seconds, one for 1 second, zero for 3 seconds, and so forth. That is, the "on" time is always 1 second, but the "off" time successively increases by 1 second between each toggle. A portion of $x(t)$ is shown in Fig. P1.1-12. Determine the energy and power of $x(t)$.

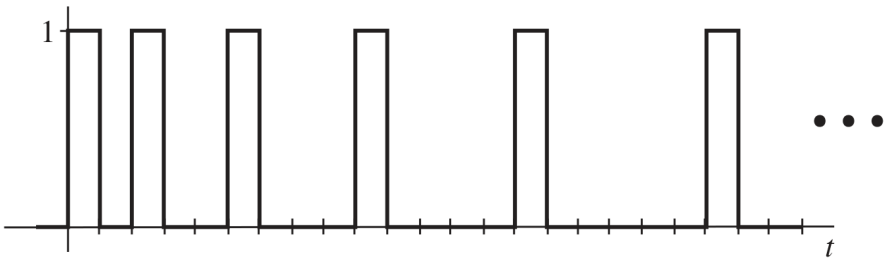


Figure P1.1-12

演算区

信号分类、能量功率与周期性的判断

Lathi 1.3-3 | 原书第 140 页

Boulet 第1章：Functions of Time as Signals, pp.2-4；Periodic Signals, p.8；Finite-Energy and Finite-Power Signals, pp.21-23

1.3-3 Determine whether each of the following statements is true or false. If the statement is false, demonstrate this by proof or example.

(a) Every continuous-time signal is an analog signal.

(b) Every discrete-time signal is a digital signal.

(c) If a signal is not an energy signal, then it must be a power signal and vice versa.

(d) An energy signal must be of finite duration.

(e) A power signal cannot be causal.

(f) A periodic signal cannot be anticausal.

演算区

由变换后图形恢复信号及奇分量

Lathi 1.5-8 | 原书第 143 页

Boulet 第1章：Transformations of the Time Variable，pp.4-8；Even and Odd Signals，pp.23-25

题干在原书左栏底部，图P1.5-8及(a)、(b)续接右栏，均已合并附上。

1.5-8 Consider the signal $y(t) = (1/5)x(-2t - 3)$ shown in Fig. P1.5-8.

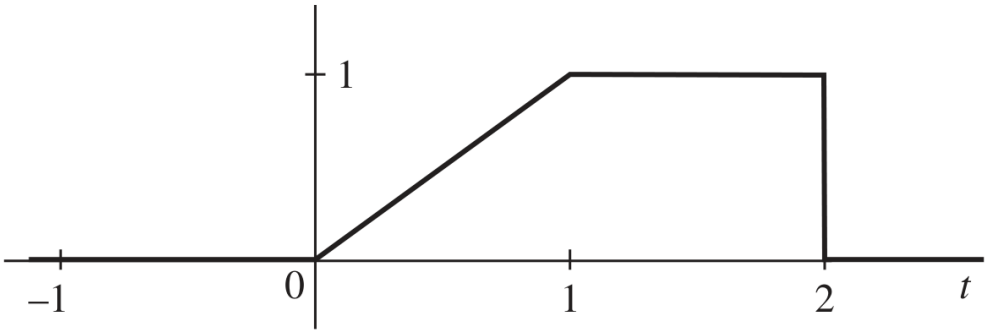


Figure P1.5-8

(a) Does $y(t)$ have an odd portion, $y_o(t)$? If so, determine and carefully sketch $y_o(t)$. Otherwise, explain why no odd portion exists.

(b) Determine and carefully sketch the original signal $x(t)$.

演算区

延时微分系统的基本性质

Lathi 1.7-8 | 原书第 145 页

Boulet 第1章：System Models and Basic Properties，pp.34-42

展开所引用1.7-4的要求：分别判断系统是否(a)BIBO稳定、(b)因果、(c)可逆、(d)线性、(e)无记忆、(f)时不变；每项回答是或否，并给出数学论证。

1.7-8 Repeat Prob. 1.7-4 for a system with input $x(t)$ that produces output $y(t)$ according to

$$y(t) = \frac{d}{dt}x(t-1)$$

演算区

有限离散序列的反转、压缩与扩展

Lathi 3.2-9 | 原书第 316 页

Boulet 第1章：Transformations of the Time Variable, pp.4-8

原题用向下箭头指定n=0样本；两幅变换图均须覆盖-5 <= n <= 4。

3.2-9 Letting $\downarrow$ identify $n = 0$, consider a DT signal $x[n]$ whose nonzero values are given as $x[n] = [1, -3, 2, 2, 3, -2, -1, 1, 2, -3, \underset{\downarrow}{3}, 3, -2, 1, -3, 2, 3, -1]$. Accurately sketch $y[n] = x[-1 - 2n]$ and $z[n] = x[-2 + n/3]$ over $-5 \leq n \leq 4$.

演算区

从阶跃响应求移位冲激与阶跃组合的输出

Lathi 2.4-37 | 原书第 231 页

Boulet 第2章：Continuous-Time LTI Systems: The Convolution Integral，pp.67-74；Properties of Linear Time-Invariant Systems: Step Response，pp.79-80

输入中的cos(sqrt(3))是与时间无关的常数，按原题保留。

2.4-37 An LTI system has step response given by $g(t) = e^{-t}u(t) - e^{-2t}u(t)$. Determine the output of this system $y(t)$ given an input $x(t) = \delta(t - \pi) - \cos(\sqrt{3})u(t)$.

演算区

左侧截断冲激响应的BIBO稳定性

Lathi 2.5-3 | 原书第 233 页

Boulet 第2章：Stability，pp.78-79

2.5-3 Consider an LTIC system with unit impulse response $h(t) = e^t\left[\frac{2}{3}\cos(\frac{3}{2}t) + \frac{1}{3}\sin(\pi t)\right] u(123 - t)$. Is this system BIBO-stable? Mathematically justify your answer.

演算区

图示有限离散序列的滑动卷积

Lathi 3.8-28 | 原书第 324 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum， pp.54-66

3.8-28 Using the sliding-tape algorithm, find $x[n] * g[n]$ for the signals shown in Fig. P3.8-28.

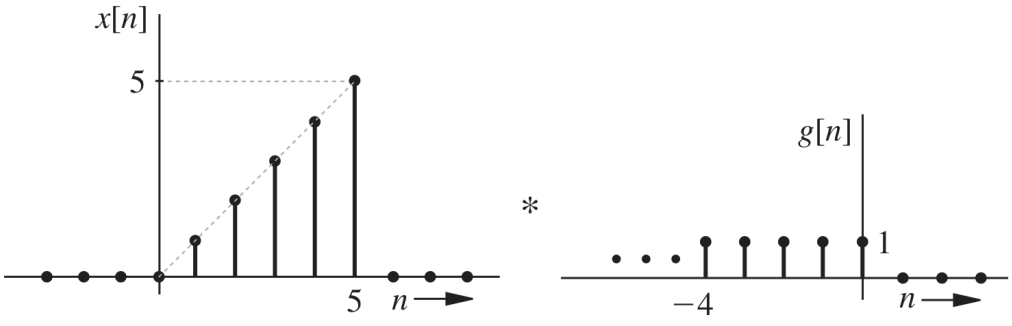


Figure P3.8-28

演算区

并联与级联系统的离散冲激响应

Lathi 3.8-34 | 原书第 325 页

Boulet 第2章：Properties of Linear Time-Invariant Systems， pp.74-80；Discrete-Time LTI Systems: The Convolution Sum， pp.54-66

完整附图P3.8-34的并联(a)与级联(b)连接方式。

3.8-34 An LTID filter has an impulse response function given by $h_1[n] = \delta[n+2] - \delta[n-2]$. A second LTID system has an impulse response function given by $h_2[n] = n(u[n+4] - u[n-4])$.

(a) Carefully sketch the functions $h_1[n]$ and $h_2[n]$ over $(-10 \leq n \leq 10)$.

(b) Assume that the two systems are connected in parallel, as shown in Fig. P3.8-34a. Determine the impulse response $h_p[n]$ for the parallel system in terms of $h_1[n]$ and $h_2[n]$. Sketch $h_p[n]$ over $(-10 \leq n \leq 10)$.

(c) Assume that the two systems are connected in cascade, as shown in Fig. P3.8-34b. Determine the impulse response $h_s[n]$ for the cascade system in terms of $h_1[n]$ and $h_2[n]$. Sketch $h_s[n]$ over $(-10 \leq n \leq 10)$.

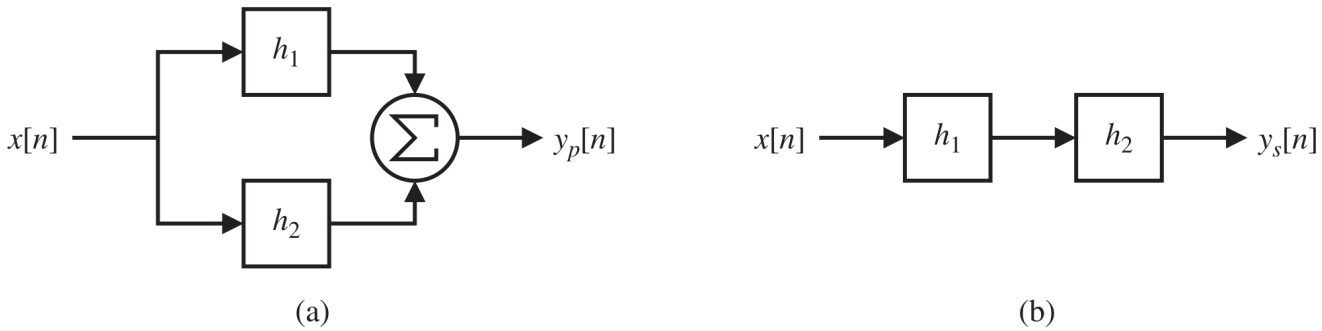


Figure P3.8-34

离散卷积在多项式系数展开中的应用

Lathi 3.8-35 | 原书第 325 页

Boulet 第2章：Discrete-Time LTI Systems: The Convolution Sum，pp.54-66

本题是有限序列卷积的代数应用；展开式为原题要求，未附结果。

3.8-35 This problem investigates an interesting application of discrete-time convolution: the expansion of certain polynomial expressions.

(a) By hand, expand $(z^3+z^2+z+1)^2$. Compare the coefficients to $[1,1,1,1]*[1,1,1,1]$.

(b) Formulate a relationship between discrete-time convolution and the expansion of constant-coefficient polynomial expressions.

(c) Use convolution to expand $(z^{-4}-2z^{-3}+3z^{-2})^4$.

(d) Use convolution to expand $(z^5+2z^4+3z^2+5)^2(z^{-4}-5z^{-2}+13)$.

演算区

双电容电路的微分模型与非零初值响应

Lathi 2.4-39 | 原书第 231 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，pp.92-96

原题元件数值、两只电容的初始电压及图P2.4-39全部保留。

2.4-39 Consider the electric circuit shown in Fig. P2.4-39.

(a) Determine the differential equation relating input $x(t)$ to output $y(t)$.

(b) Determine the output $y(t)$ in response to the input $x(t) = 4te^{-3t/2}u(t)$. Assume component values of $R = 1\ \Omega$, $C_1 = 1$ F, and $C_2 = 2$ F, and initial capacitor voltages of $V_{C_1} = 2$ V and $V_{C_2} = 1$ V.

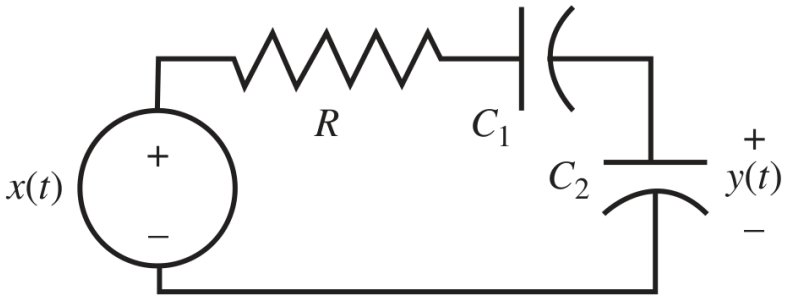


Figure P2.4-39

演算区

理想运放积分电路的微分模型与阶跃响应

Lathi 2.4-44 | 原书第 232 页

Boulet 第3章：Causal LTI Systems Described by Differential Equations，pp.92-96；Characteristic Polynomials and Stability，pp.112-116

2.4-44 Consider the circuit shown in Fig. P2.4-44. This circuit functions as an integrator. Assume ideal op-amp behavior and recall that

$$i_C(t) = C\frac{dV_C(t)}{dt}$$

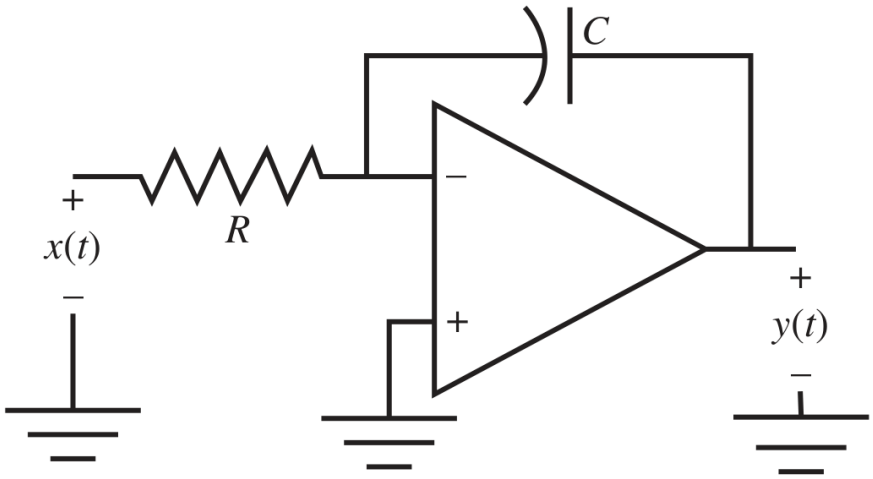


Figure P2.4-44

(a) Determine the differential equation that relates the input $x(t)$ to the output $y(t)$.

(b) This circuit does not behave well at dc. Demonstrate this by computing the zero-state response $y(t)$ for a unit step input $x(t) = u(t)$.

演算区

由离散冲激响应构造差分方程并判断性质

Lathi 3.4-6 | 原书第 317 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations， pp.96-101；Impulse Response of a Difference LTI System， pp.109-112；结合第1章系统性质

保留原序列的n=0箭头及(a)至(f)全部要求。

3.4-6 Letting $\downarrow$ identify $n = 0$, define the nonzero values of signal $g[n]$ in vector form as $[1, 2, 3, 4, 5, 4, 3, \underset{\downarrow}{2}, 1]$. The impulse response of an LTID system is defined in terms of $g[n]$ as $h[n] = g[-2n - 1]$.

(a) Express the nonzero values of $h[n]$ in vector form, taking care to identify the $n = 0$ point.

(b) Write a constant-coefficient linear difference equation (input $x[n]$ and output $y[n]$) that has impulse response $h[n]$.

(c) Show that the system is both linear and time-invariant.

(d) Determine, if possible, whether the system is BIBO-stable.

(e) Determine, if possible, whether the system is memoryless.

(f) Determine, if possible, whether the system is causal.

演算区

二阶销售估计差分方程的递推求解

Lathi 3.5-4 | 原书第 319、255 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations，pp.96-101

附上原题引用的式(3.6)，仅作为已知方程；求前三项，初值和输入按原题给定，不附例题解答。

3.5-4 Solve recursively the second-order difference Eq. (3.6) for sales estimate (first three terms only), assuming $y[-1] = y[-2] = 0$ and $x[n] = 100u[n]$.

$$y[n+2] + \tfrac{1}{4}y[n+1] + \tfrac{1}{16}y[n] = x[n+2] \tag{3.6}$$

演算区

咖啡补充过程的差分模型、总响应与稳态

Lathi 3.8-36 | 原书第 325、326 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations，pp.96-101；Characteristic Polynomials and Stability，pp.112-116

原题跨页叙述与(a)至(d)全部接续，初始干净杯条件及最后的输入改进要求完整保留。

3.8-36 Joe likes coffee, and he drinks his coffee according to a very particular routine. He begins by adding two teaspoons of sugar to his mug, which he then fills to the brim with hot coffee. He drinks 2/3 of the mug's contents, adds another two teaspoons of sugar, and tops the mug off with steaming hot coffee. This refill procedure continues, sometimes for many, many cups of coffee. Joe has noted that his coffee tends to taste sweeter with the number of refills.

Let independent variable n designate the coffee refill number. In this way, $n = 0$ indicates the first cup of coffee, $n = 1$ is the first refill, and so forth. Let $x[n]$ represent the sugar (measured in teaspoons) added into the system (a coffee mug) on refill n. Let $y[n]$ designate the amount of sugar (again, teaspoons) contained in the mug on refill n.

演算区

咖啡补充过程的差分模型、总响应与稳态（续）

Lathi 3.8-36 | 原书第 325、326 页

Boulet 第3章：Causal LTI Systems Described by Difference Equations，pp.96-101；Characteristic Polynomials and Stability，pp.112-116

原题跨页叙述与(a)至(d)全部接续，初始干净杯条件及最后的输入改进要求完整保留。

(a) The sugar (teaspoons) in Joe's coffee can be represented using a standard second-order constant coefficient difference equation $y[n] + a_1 y[n-1] + a_2 y[n-2] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2]$. Determine the constants a_1, a_2, b_0, b_1, and b_2.

(b) Determine $x[n]$, the driving function to this system.

(c) Solve the difference equation for $y[n]$. This requires finding the total solution. Joe always starts with a clean mug from the dishwasher, so $y[-1]$ (the sugar content before the first cup) is zero.

(d) Determine the steady-state value of $y[n]$. That is, what is $y[n]$ as $n \to \infty$? If possible, suggest a way of modifying $x[n]$ so that the sugar content of Joe's coffee remains a constant for all nonnegative n.

演算区

（复习）多谐波三角级数与复指数频谱

Lathi 6.3-4 | 原书第 672 页

Boulet 第4章：Determination of the Fourier Series Representation，pp.134-136；Graph of the Fourier Series Coefficients: The Line Spectrum，pp.137-138

复习题：此前见第4批。复习第4批：重练三角式、幅相谱与复指数式的相互转换，特别留意直流项、负系数和相位；(a)-(d)完整保留。

6.3-4 The trigonometric Fourier series of a certain periodic signal is given by

$$x(t) = 3 + \sqrt{3}\cos 2t + \sin 2t + \sin 3t - \frac{1}{2}\cos\left(5t + \frac{\pi}{3}\right)$$

(a) Sketch the trigonometric Fourier spectra.
(b) By inspection of spectra in part (a), sketch the exponential Fourier series spectra.
(c) By inspection of spectra in part (b), write the exponential Fourier series for $x(t)$.
(d) Show that the series found in part (c) is equivalent to the trigonometric series for $x(t)$.

演算区

（复习）由三角幅相谱重建信号并转为复指数谱

Lathi 6.3-6 | 原书第 673 页

Boulet 第4章：Determination of the Fourier Series Representation，pp.135-136；Graph of the Fourier Series Coefficients: The Line Spectrum，pp.137-138

复习题：此前见第4批。复习第4批：重练读谱及两种傅里叶级数之间的转换。原图 P6.3-6 的幅度谱、相位谱、坐标标记与(a)(b)标签均已附。

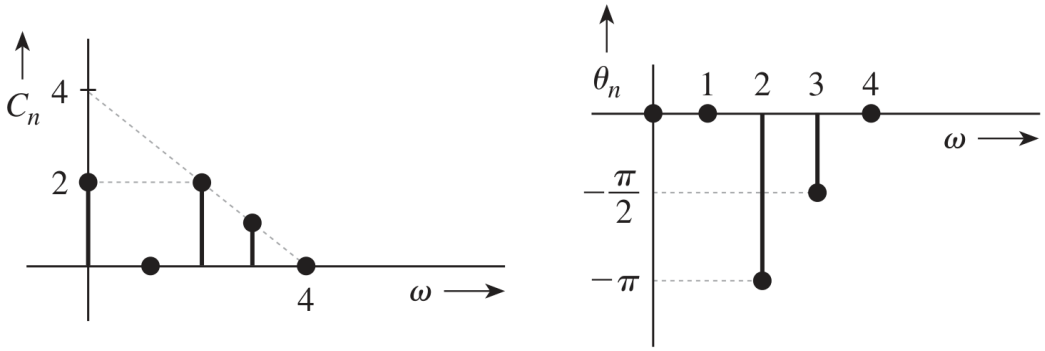
6.3-6 Figure P6.3-6 shows the trigonometric Fourier spectra of a periodic signal $x(t)$.

(a) By inspection of Fig. P6.3-6, find the trigonometric Fourier series representing $x(t)$.

(b) By inspection of Fig. P6.3-6, sketch the exponential Fourier spectra of $x(t)$.

(c) By inspection of the exponential Fourier spectra obtained in part (b), find the exponential Fourier series for $x(t)$.

(d) Show that the series found in parts (a) and (c) are equivalent.



(a) (b)

Figure P6.3-6

演算区

（复习）分段绝对值周期信号的傅里叶级数

Lathi 6.3-16 | 原书第 675、635 页

Boulet 第4章：Determination of the Fourier Series Representation，pp.134-135；Properties of Continuous-Time Fourier Series，pp.139-140（性质法拓展）

复习题：此前见第4批。复习第4批：重练分段周期信号的表示、直流项与性质法。题目同页跨栏，(a)-(d)均已附；例6.11仅为方法参考，补附原书表6.2供查性质，不附例题解答。

6.3-16 A 3-periodic signal $x(t)$ is defined as

$$x(t) = \begin{cases} |t| & -1 \le t \le 1 \\ 0 & 1 < |t| \le 1.5 \\ x(t+3) & \forall t \end{cases}$$

(a) Plot $x(t)$ over $-3 \le t \le 3$.

(b) Determine D_0, the dc content of $x(t)$.

(c) Similar to Ex. 6.11, use properties and not integration to determine D_n for $n \neq 0$.

(d) Plot the magnitude spectrum $|D_n|$ over a suitable range of n. What is the most dominant frequency component of this signal?

TABLE 6.2 Selected Fourier Series Properties

Operation	$\boldsymbol{x(t)}$	$\boldsymbol{D_n}$
Scalar multiplication	$kx(t)$	kD_n
Addition	$x_1(t) + x_2(t)$ $x_1(t), x_2(t)$ require same ω_0	$D_{1,n} + D_{2,n}$
Conjugation	$x^*(t)$	D^*_{-n}
Reversal	$x(-t)$	D_{-n}
Time shifting	$x(t - t_0)$	$D_n e^{-jn\omega_0 t_0}$
Frequency shifting	$x(t)e^{jn_0\omega_0 t}$	D_{n-n_0}
Frequency convolution	$x_1(t)x_2(t)$ $x_1(t), x_2(t)$ require same ω_0	$D_{1,n} * D_{2,n}$
Time differentiation	$\dfrac{d^k x(t)}{dt^k}$	$(jn\omega_0)^k D_n$

（复习）周期指数输入经过RC系统的稳态响应

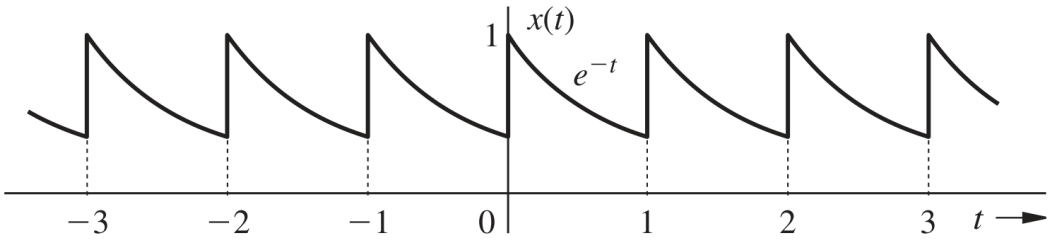
Lathi 6.4-5 | 原书第 676 页

Boulet 第4章：Steady-State Response of an LTI System to a Periodic Signal，pp.155-156

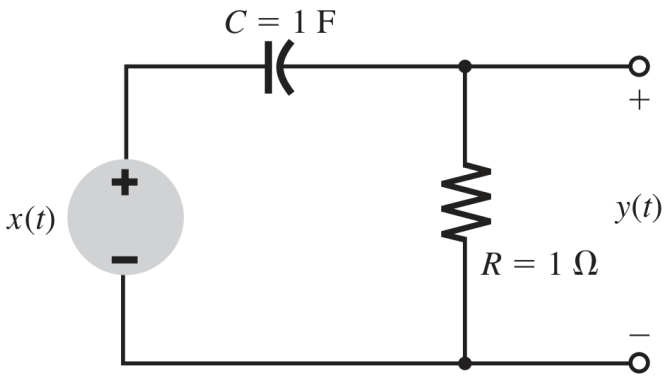
复习题：此前见第4批。复习第4批：重练由周期输入的傅里叶级数求系统稳态响应。原图 P6.4-5(a)的输入和(b)的RC电路均已附；核对输入、输出端与元件位置，图中 C=1 F、R=1 Ω。

6.4-5 (a) Find the exponential Fourier series for a periodic signal $x(t)$ shown in Fig. P6.4-5a.

(b) The signal $x(t)$ is applied at the input of an LTIC system shown in Fig. P6.4-5b. Find the expression for the output $y(t)$.



(a)



(b)

Figure P6.4-5

演算区

（复习）正交信号的能量可加性与向量类比

Lathi 6.5-3 | 原书第 677 页

Boulet 第4章：Optimality and Convergence of the Fourier Series，pp.144-145；Parseval Theorem，p.150（正交分量的能量可加性）

复习题：此前见第4批。复习第4批：重练正交性、能量分解与向量类比，巩固傅里叶级数正交展开的基础；题中的能量等式是原题要求证明的结论。

6.5-3 If $x(t)$ and $y(t)$ are orthogonal, then show that the energy of the signal $x(t)+y(t)$ is identical to the energy of the signal $x(t)-y(t)$ and is given by E_x+E_y. Explain this result by using the vector analogy. In general, show that for orthogonal signals $x(t)$ and $y(t)$ and for any pair of arbitrary real constants c_1 and c_2, the energies of $c_1x(t)+c_2y(t)$ and $c_1x(t)-c_2y(t)$ are both given by $c_1^2E_x+c_2^2E_y$.

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